

Q.157 | LC - 2095: Delete Middle Node of a Linked List

Apple

Bloomberg

This question can be done in 2 passes or even a single pass.

Approach 1: Two Pass Solution

- **S1:** Find the length of the Linked List.
- **S2:** Calculate the middle point:

```
mid = len / 2
```

- **S3:** Traverse to the node to (delete - 1), because we need to reestablish the connection.

```
ListNode* dummy = new ListNode(0, head); ListNode* curr =  
dummy; for (int i = 0; i < mid; i++) { curr = curr->next; }  
ListNode* tod = curr->next; curr->next = curr->next->next;  
delete (tod); ListNode* ans = dummy->next; delete (dummy);  
return ans;
```

Approach 2: Two Pointer (Fast and Slow)

- **S1:** When the fast pointer reaches the end, the slow pointer reaches the mid node.

- **S2:** However, since we need one node *prior* to the mid node (to delete the mid node), we use a dummy node such that:
 - ◦ `slow` starts from dummy
 - ◦ `fast` starts from head
- **S3:** Reestablish the connection and delete the target node.